Module - 3

Correlation and regression:

Correlation:

Correlation is a statistical tool that helps to measure and analyze the relationship between two variables.

The measure of correlation called correlation coefficient.

Methods of studying correlation:

a) Scatter diagram
b) Karl pearson's coefficient of correlation
c) Spearman's Rank correlation coefficient
d) Method of least squares.

Types of Correlation:

Type 1:

Positive Correlation:

The Correlation is Said to be positive Correlation if the Values of two variables Changeing with Same direction.

As X is Increasing, Y is increasing
As X is decreasing, Y is decreasing

Ex: As height increases, So does weight.

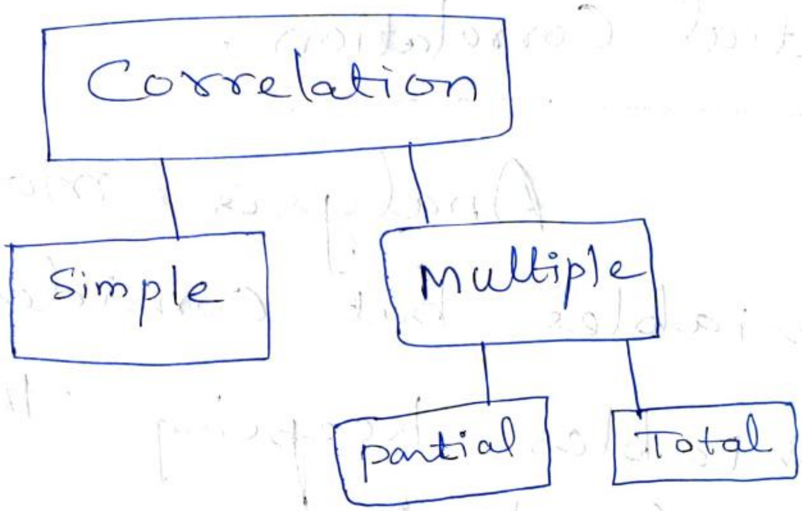
Negative Correlation:

The Correlation is Said to be negative Correlation when the values of variables Change with opposite direction.

As X is increasing, Y is decreasing
As X is decreasing, Y is increasing

Ex: As TV time increases, grades decrease.

Type 2:



Simple Correlation:

Under Simple Correlation there are only two Variables are Studied.

Multiple Correlation:

Under multiple Correlation three or more than three Variables are Studied.

Partial Correlation.

Analysis more than two Variables but consider only two variables keeping the other variables as Constant.

Total Correlation:

It is based on all the relevant Variables, which is normally not feasible.

Type-III

Linear Correlation:

Correlation is said to be linear when the amount of change in one variable tends to bear a constant ratio to the amount of change in the other.

The graph of the variables having a linear relationship will form a straight line.

Ex: $X = 1, 2, 3, 4, 5, 6, 7, 8$

$Y = 5, 7, 9, 11, 13, 15, 17, 19$

$Y = 3 + 2X.$

Non Linear Correlation:

The Correlation is said to be non linear if the amount of change in one variable does not bear a constant ratio to the amount of change in the other variable.

Scatter Diagram Method:

Scatter Diagram is a graph of observed plotted points where each points represents the values of X & Y as a coordinate. It portrays the relationship between these two variables graphically.

A perfect positive correlation.

r = 1

Positive relationship

r = 0.80

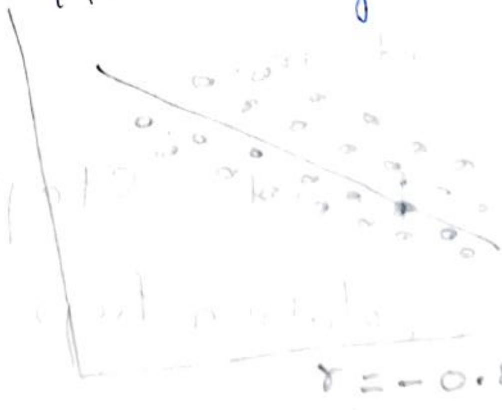
Moderate positive correlation

r = 0.4

Perfect negative correlation

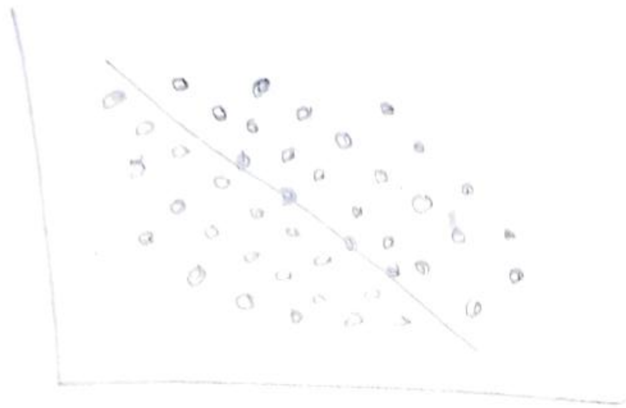
r = -1.0

Moderate negative correlation



r = -0.80

Weak negative Correlation



$r = -0.2$

No Correlation

$r = 0$

Advantages of Scatter Diagram:

Simple & Non-Mathematical method

Not influenced by the size of extreme item.

First step in investing the relationship between two variables.

Disadvantage of Scatter diagram

Can not adopt the an exact degree of correlation.

Univariate distribution:

The distribution involving only one variable.

Bivariate distribution:

The distribution involving two variables.

Karl-pearson's Coefficient of correlation

As a measure of itensity or degree of linear relationship between two variables, Karl Pearson (1867-1936), a British Biometrician, developed a formula called Correlation Coefficient

Correlation Coefficient between two random variables X and Y, usually denoted by $r(X,Y)$ or r_{XY}, is a numberical measure of linear relationship between them and is defined as

$$r(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Where $Cov(X,Y) = \frac{1}{n}\Sigma XY - \bar{X}\bar{Y}$

$$\sigma_X = \sqrt{\frac{1}{n}\Sigma X^2 - \bar{X}^2}$$

$$\sigma_Y = \sqrt{\frac{1}{n}\Sigma Y^2 - \bar{Y}^2}$$

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$$\sigma_x = \sqrt{\frac{1}{n}\Sigma x^2 - \bar{x}^2}$$

$$\sigma_y = \sqrt{\frac{1}{n}\Sigma y^2 - \bar{y}^2}$$

Where n is the number of items in the given data

Limits of Correlation Coefficient:

We know that

$$r(x,y) = \frac{\operatorname{cov}(x,y)}{\sigma_x \sigma_y}$$

$$= \frac{\frac{1}{n}\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\frac{1}{n}\Sigma(x_i-\bar{x})^2 \frac{1}{n}\Sigma(y_i-\bar{y})^2}}$$

$$r(x,y) = \frac{\Sigma a_i b_i}{\sqrt{\Sigma a_i^2 \, \Sigma b_i^2}}$$

where, $a_i = x_i - \bar{x}$

$b_i = y_i - \bar{y}$

Squaring on both sides, we get

$$r^2(x,y) = \frac{(\Sigma a_i b_i)^2}{\Sigma a_i^2 \, \Sigma b_i^2} \longrightarrow (1)$$

By schwartz inequality,

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)$$

ie $$\frac{\left(\sum_{i=1}^{n} a_i b_i\right)^2}{\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)} \le 1 \longrightarrow (2)$$

Using (2) in (1), we get

$$\therefore\ r^2(x,y) \le 1$$

$$\therefore\ |r(x,y)| \le 1$$

$$\Longrightarrow \boxed{-1 \le r \le 1}$$

Note:

1. Correlation Coefficient does not exceed unity.

2. Two independent variables are uncorrelated since $Cov(x,y) = 0$ when X & Y are independent variables.

$$\therefore\ r(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = 0.$$

(1) Calculate the correlation Coefficient for the following heights (in inches) of fathers X and their sons Y.

X :	65	66	67	67	68	69	70	72
Y :	67	68	65	68	72	72	69	71

Solution: Method-1:

X	Y	XY	X^2	Y^2
65	67	4355	4225	4489
66	68	4488	4356	4624
67	65	4355	4489	4225
67	68	4556	4489	4624
68	72	4896	4624	5184
69	72	4968	4761	5184
70	69	4830	4900	4761
72	71	5112	5184	5041
544	552	37560	37028	38132

Here $n=8$

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\bar{x} = \frac{544}{8} = 68$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{552}{8} = 69$$

$$\bar{x}\bar{y} = 68 \times 69 = 4692$$

$$\sigma_x = \sqrt{\frac{1}{n}\Sigma x^2 - \bar{x}^2}$$

$$= \sqrt{\frac{37028}{8} - 4624}$$

$$\sigma_x = 2.121$$

$$\sigma_y = \sqrt{\frac{1}{n}\Sigma y^2 - \bar{y}^2}$$

$$= \sqrt{\frac{38132}{8} - 4761}$$

$$\sigma_y = 2.345$$

$$\therefore r(x,y) = \frac{\frac{1}{n}\Sigma xy - \bar{x}\bar{y}}{\sigma_x \sigma_y} = \frac{\frac{1}{8}37560 - 4692}{2.121 \times 2.345}$$

$$= 0.6030$$

Note:

Correlation Coefficient is independent of change of origin and scale.

ie $r(x,y) = r(u,v)$

where $u = \frac{x-a}{h}$, $v = \frac{y-b}{k}$

Where a and b are some arbitrary constants usually the mid values of the given data x & y respectively, and h & k are constants, $h>0$ & $k>0$.

Method-2:

x	y	$u=x-68$	$v=y-68$	uv	u^2	v^2
65	67	-3	-1	3	9	1
66	68	-2	0	0	4	0
67	65	-1	-3	3	1	9
67	68	-1	0	0	1	0
68	72	0	4	0	0	16
69	72	1	4	4	1	16
70	69	2	1	2	4	1
72	71	4	3	12	16	9
		0	8	24	36	52

$$\bar{u} = \frac{\Sigma u}{n} = \frac{0}{8} = 0$$

$$\bar{v} = \frac{\Sigma v}{n} = \frac{8}{8} = 1$$

$$Cov(x,y) = Cov(u,v)$$

$$= \frac{\Sigma uv}{n} - \bar{u}\bar{v}$$

$$= \frac{24}{8} - 0$$

$$= 3$$

$$\sigma_u = \sqrt{\frac{\Sigma u^2}{n} - \bar{u}^2} = \sqrt{\frac{36}{8} - 0} = 2.121$$

$$\sigma_v = \sqrt{\frac{\Sigma v^2}{n} - \bar{v}^2} = \sqrt{\frac{52}{8} - 1} = 2.345$$

$$r(x,y) = r(u,v) = \frac{Cov(u,v)}{\sigma_u \sigma_v} = \frac{3}{2.121 \times 2.345}$$

$$= 0.6031$$

Note: This method is very useful and time saving.

(2) Let X, Y and Z be uncorrelated random variables with zero means and standard deviations 5, 12 and 9 respectively. If $U = X+Y$ and $V = Y+Z$, find the correlation coefficient between U and V.

Sol:

Given that all the three random variables have zero mean. Hence

$E(X) = E(Y) = E(Z) = 0$.

Now, $Var(X) = E(X^2) - (E(X))^2$

$E(X^2) = Var(X)$

$= 5^2$

$E(X^2) = 25$

S.D $= \sqrt{\text{Variance}}$

Variance $= (S.D)^2$

|||^{ly} $E(Y^2) = 12^2 = 144$

$E(Z^2) = 9^2 = 81$

Since X & Y are uncorrelated we have $Cov(X, Y) = 0$

$\Rightarrow E(XY) = E(X)E(Y)$

$E(XY) = 0$.

$|||^{ly}$ $E(YZ) = 0$

$E(ZX) = 0$

$$r(U, V) = \frac{E(UV) - E(U)E(V)}{\sigma_U \sigma_V}$$

Now, $E(U) = E(X+Y)$

$= E(X) + E(Y)$

$= 0$

$E(V) = E(Y+Z)$

$= E(Y) + E(Z)$

$E(V) = 0$

$$E(U^2) = E((x+y)^2)$$

$$= E(x^2) + E(y^2) + 2E(xy)$$

$$= 25 + 144 + 0$$

$$E(U^2) = 169$$

$$E(V^2) = E((y+z)^2)$$

$$= E(y^2) + E(z^2) + 2E(yz)$$

$$= 144 + 81 + 0$$

$$E(V^2) = 225$$

$$\text{var}(U) = E(U^2) - (E(U))^2$$

$$\text{var}(U) = 169 - 0 = 169$$

$$\sigma_U = \sqrt{169} = 13$$

$$\text{var}(V) = E(V^2) - (E(V))^2 = 225 - 0 = 225$$

$$\sigma_V = \sqrt{225} = 15$$

$$E(UV) = E((X+Y)(Y+Z))$$

$$= E(XY) + E(Y^2) + E(XZ) + E(YZ)$$

$$= E(Y^2)$$

$$E(UV) = 144$$

$$\therefore \ r(U,V) = \frac{144-0}{13 \times 15}$$

$$r(U,V) = \frac{144}{195} = \frac{48}{65}$$

(3) The independent variables X and Y have the probability density function given by

$$f_x(x) = \begin{cases} 4ax, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$f_y(y) = \begin{cases} 4by, & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the correlation coefficient between X+Y and X−Y.

Sol:

Let $U = X+Y$, $V = X-Y$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^1 x \, 4ax \, dx$$

$$E(x) = 4a \left(\frac{x^3}{3} \right)_0^1 = \frac{4a}{3}$$

$|||^{ly}$ $E(Y) = \frac{4b}{3}$

Since X & Y are independent, the joint pdf of X and Y is given by

$$f(x,y) = f(x) f(y)$$

$$= (4ax)(4by)$$

$$= 16abxy \ , \quad 0 \leq x \leq 1, \ 0 \leq y \leq 1.$$

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\, dx\, dy$$

$$= \int_0^1\int_0^1 xy\,(16abxy)\, dx\, dy$$

$$= 16ab \int_0^1\int_0^1 x^2y^2\, dx\, dy$$

$$= 16ab \int_0^1 \left(\frac{x^3}{3} y^2\right)_0^1 dy$$

$$= \frac{16ab}{3}\int_0^1 y^2\, dy$$

$$= \frac{16ab}{3}\left(\frac{y^3}{3}\right)_0^1$$

$$= \frac{16ab}{9}$$

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$= \frac{16ab}{9} - \left(\frac{4a}{3}\right)\left(\frac{4b}{3}\right)$$

$$= \frac{16ab}{9} - \frac{16ab}{9}$$

$$= 0.$$

$\therefore \ r(x, y) = 0$.

Note: 1. Since $Cov(x, y) = 0$ we need not Calculate the σ_x & σ_y.

2. Since $Cov(x, y) = 0$, X & Y are uncorrelated.

Rank Correlation Coefficient :

Sometimes the actual numeri[c] values of X & Y may not be avail[able] but the positions of the actual values arranged in order of merit (ranks) only may be available. The ranks of X & Y will in gen[eral] be different.

For example, if we consider the relatio[n] between intelligence & beauty, it is not necessary that a beautiful individ[ual] is intelligent also.

The rank correlation coefficient between X & Y denoted by $\rho(X, Y)$ and is defined by

$$\rho(X, Y) = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

where $d_i = x_i - y_i$

This formula is called Spearman's formula for rank correlation.

Note:

$\Sigma d_i = \Sigma (x_i - y_i) = \Sigma x_i - \Sigma y_i = 0$ always

(1) Find the rank Correlation Coefficient from the following data.

Rank in X	1	2	3	4	5	6	7
Rank in y	4	3	1	2	6	5	7

Sol:

X	y	$d_i = x_i - y_i$	d_i^2
1	4	−3	9
2	3	−1	1
3	1	2	4
4	2	2	4
5	6	−1	1
6	5	1	1
7	7	0	0
		0	20

Rank corelation Coefficient

$$\rho_{xy} = 1 - \frac{6 \Sigma d_i^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 20}{7(49-1)}$$

$$= 1 - \frac{120}{336} = 0.6429.$$

(2) The ranks of some 16 students in Mathematics and Physics are as follows. Calculate rank correlation coefficients for proficiency in mathematics & physics.

Ranks in Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Ranks in physics	1	10	3	4	5	7	2	6	8	11	15	9	14	12	16	13

Ans: $\rho = 0.8$ //

(3) 10 Competitors in a musical test were ranked by 3 judges x, y, z in the following order.

	A	B	C	D	E	F	G	H	I	J
Rank by x	1	6	5	10	3	2	4	9	7	8
Rank by y	3	5	8	4	7	10	2	1	6	9
Rank by z	6	4	9	8	1	2	3	10	5	7

using rank correlation method, discuss which pair of judges has the nearest approach to common likings of music.

Sol:

x_i	y_i	z_i	$d_1 = x_i - y_i$	$d_2 = y_i - z_i$	$d_3 = x_i - z_i$	d_1^2	d_2^2	d_3^2
1	3	6	-2	-3	-5	4	9	25
6	5	4	$+1$	1	2	1	1	4
5	8	9	-3	-1	-4	9	1	16
10	4	8	6	-4	2	36	16	4
3	7	1	-4	6	2	16	36	4
2	10	2	-8	8	0	64	64	0
4	2	3	2	-1	1	4	1	1
9	1	10	8	-9	-1	64	81	1
7	6	5	1	1	2	1	1	4
8	9	7	-1	2	1	1	4	1
						200	214	60

$$\rho_1(x, y) = 1 - \frac{6 \Sigma d_1^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 200}{10(100-1)} = -0.212$$

$$\rho_2(y, z) = 1 - \frac{6 \Sigma d_2^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 214}{10(10^2-1)} = -0.296$$

$$\rho_3(x, z) = 1 - \frac{6 \Sigma d_3^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 60}{10(100-1)}$$

$$= 0.636$$

Since the rank Correlation between x & z is maximum and also positive, we Conclude that the pair of judges x and z has the nearest approach to common likings in music.

Repeated ranks:

In the correlation formula, we add the factor $\frac{m(m^2-1)}{12}$ to Σd^2 where m is the number of items is repeated. This correlation factor is to be added for each repeated value.

(1) Obtain the rank correlation coefficient for the following data.

X : 68 64 75 50 64 80 75 40 55 64

Y : 62 58 68 45 81 60 68 48 50 70

Sol:

X	Y	Rank X x_i	Rank Y y_i	$d_i = x_i - y_i$	d_i^2
68	62	4	5	-1	1
64	58	6	7	-1	1
75	68	2.5	3.5	-1	1
50	45	9	10	-1	1
64	81	6	1	5	25
80	60	1	6	-5	25
75	68	2.5	3.5	-1	1
40	48	10	9	1	1
55	50	8	8	0	0
64	70	6	2	4	16
					72

In x series 75 repeated twice which are in the positions 2nd and 3rd ranks. Therefore common ranks 2.5 (which is the average of 2 and 3) is to be given for each 75.

Also in x series 64 is repeated thrice which are in the position 5th, 6th and 7th ranks. Therefore common ranks 6 (which is the average of 5, 6 and 7) is to be given for each 64.

Similarly in y series 68 is repeated twice which are in the positions 3rd and 4th ranks. Therefore common ranks 3.5 (which is the average of 3 and 4) is to be given for each 68.

Correlation Factors:

In X series 75 repeated twice

$\therefore$ $C.F = \frac{2(2^2-1)}{12} = \frac{1}{2}$

In X series 64 repeated thrice

$\therefore$ $C.F = \frac{3(3^2-1)}{12} = \frac{24}{12} = 2$

In Y series 68 repeated twice

$\therefore$ $C.F = \frac{2(2^2-1)}{12} = \frac{1}{2}$

$\therefore$ Rank Correlation $\rho = 1 - \frac{6(\Sigma d^2 + \frac{1}{2} + 2 + \frac{1}{2})}{10(10^2-1)}$

$= 1 - \frac{6(72+3)}{10(99)}$

$= 1 - \frac{450}{990}$

$= 0.5454.$

H.W

A Sample of 12 fathers and their eldest Sons have the following data about their heights in inches.

Fathers :	65	63	67	64	68	62	70	66	68	67	69	71
Sons	68	66	68	65	69	66	68	65	71	67	68	70

Calculate the rank Correlation Coefficient.

Ans: $\rho = 0.722$.

Regression Analysis:

Regression is the average relationship between two Variables.

Regression Analysis is mathematical measure of average relationship between two or more Variables.

Regression analysis is a Satistical tool used in prediction of value of unknown variable from known variable

Lines of Regression:

If the variables in a bivariate distribution are related we will find that the points in the scattered diagram will cluster around some curve called the curve of regression. If the curve is a straight line, it is called the line of regression and there is said to be linear regression between the variables, otherwise regression is said to be curvillinear.

The line of regression of y on x is given by

$$y-\bar{y} = r\frac{\sigma_y}{\sigma_x}(x-\bar{x})$$

where r is the correlation coefficient σ_y and σ_x are standard deviations.

The line of regression of x on y is given by

$$x-\bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y}).$$

Note:

Both the lines of regression passes through $(\bar{x}, \bar{y})$.

Angle between two lines of regress.

If the equations of lines of regression of y on x and x on y are

$$y-\bar{y} = r\frac{\sigma_y}{\sigma_x}(x-\bar{x})$$

and

$$x-\bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y}).$$

The angle θ between the two lines of regression is given by

$$\theta = \tan^{-1}\left(\frac{1-r^2}{r}\left(\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}\right)\right)$$

Note:

1. If $r=0$, we get $\tan\theta = \infty$

$$\Rightarrow \theta = \pi/2$$

$\therefore$ when $r=0$, the lines of regression are perpendicular to each other.

2. If $r = \pm 1$, then

$$\tan\theta = 0$$

$$\Rightarrow \theta = 0 \text{ or } \pi.$$

$\therefore$ When $r = \pm 1$, the two regression lines are parallel to each other (or) coincide.

Regression Coefficients:

Regression Coefficient of y on x

$$r\frac{\sigma_y}{\sigma_x} = b_{yx} \longrightarrow (1)$$

Regression Coefficient of x on y

$$r\frac{\sigma_x}{\sigma_y} = b_{xy} \longrightarrow (2)$$

$$(1)\times(2) \Rightarrow r^2 = b_{yx}\times b_{xy}$$

$$\therefore r = \pm\sqrt{b_{yx}\times b_{xy}}$$

ie correlation coefficient $r = \pm\sqrt{b_{yx}\times b_{xy}}$.

Note:

1. $$b_{yx} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}$$

$$b_{xy} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2}$$

2. Regression Coefficients are independent of change of origin but not of scale.

ie $b_{yx} = b_{vu}$

$$b_{yx} = \frac{n\Sigma uv - (\Sigma u)(\Sigma v)}{n\Sigma u^2 - (\Sigma u)^2}$$

and $b_{xy} = b_{uv}$

$$= \frac{n\Sigma uv - (\Sigma u)(\Sigma v)}{n\Sigma v^2 - (\Sigma v)^2}$$

where $u = x-a$, $v = y-b$.

(1) From the following data, find

(i) The two regression equations.

(ii) The Coefficient of correlation between the marks in Economics & Statistics.

(iii) The most likely marks in Statistics when marks in Economics are 30.

Marks in Economics	25	28	35	32	31	36	29	38	34	32
Marks in Statistics	43	46	49	41	36	32	31	30	33	39

Sol:

x	y	$x-\bar{x}$ $x-32$	$y-\bar{y}$ $y-38$	$(x-\bar{x})^2$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
25	43	-7	5	49	25	-35
28	46	-4	8	16	64	-32
35	49	3	11	9	121	33
32	41	0	3	0	9	0
31	36	-1	-2	1	4	2
36	32	4	-6	16	36	-24
29	31	-3	-7	9	49	21
38	30	6	-8	36	64	-48
34	33	2	-5	4	25	-10
32	39	0	1	0	1	0
320	380	0	0	140	398	-93

$$\bar{x} = \frac{\Sigma x}{n} = \frac{320}{10} = 32$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{380}{10} = 38$$

Coefficient of regression of y on x is

$$b_{yx} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}$$

$$= \frac{-93}{140} = -0.6643$$

Coefficient of regression of x on y is

$$b_{xy} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2}$$

$$= \frac{-93}{398} = -0.2337$$

Equation of the line of regression of x on y is

$$x - \bar{x} = b_{xy}(y-\bar{y})$$

$$x - 32 = -0.2337(y - 38)$$

$$x - 32 = -0.2337y + 0.2337 \times 38$$

$$x = -0.2337y + 40.8806$$

Equation of the line of regression of y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 38 = -0.6643(x - 32)$$

$$y = -0.6643x + 0.6643 \times 32 + 38$$

$$y = -0.6643x + 59.2576$$

Coefficient of correlation

$$r^2 = b_{yx} \times b_{xy}$$

$$= -0.6643 \times (-0.2337)$$

$$r^2 = 0.1552$$

$$r = \pm 0.394.$$

Now we have to find the most likely marks in Statistics (y) when marks in Economics (x) are 30.

We use the line of regression of y on x

ie $y = -0.6643x + 59.2576$

$= -0.6643 \times 30 + 59.2576$

$y = 39.3286$

$y \simeq 39$.

) Height of fathers and Sons are given in centimeters.

x : Height of father	150	152	155	157	160	161	164	166
y : Height of Son	154	156	158	159	160	162	161	164

Find the two lines of regression and calculate the expected average height of the son when the height of the father is 154 cm.

x	y	$u = x-160$	$v = y-159$	u^2	v^2	uv
150	154	-10	-5	100	25	50
152	156	-8	-3	64	9	24
155	158	-5	-1	25	1	5
157	159	-3	0	9	0	0
160	160	0	1	0	1	0
161	162	1	3	1	9	3
164	161	4	2	16	4	8
166	164	6	5	36	25	30
1265	1274	-15	2	251	74	120

$$\bar{x} = \frac{\Sigma x}{n} = \frac{1265}{8} = 158.125$$

(or)

$$\bar{x} = A + \frac{\Sigma u}{n}$$ where A is assumed mean.

$$= 160 - \frac{15}{8}$$

$$= 158.125.$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{1274}{8} = 159.25$$

(or) $$\bar{y} = A + \frac{\Sigma v}{n} = 159 + 2/8 = 159.2$$

Since regression Coefficients are indep of change of origin, we have regres Coefficient of y on x is

$$b_{yx} = b_{vu} = \frac{n\Sigma uv - (\Sigma u)(\Sigma v)}{n\Sigma u^2 - (\Sigma u)^2}$$

$$= \frac{8 \times 120 - (-15)(2)}{8(251) - (-15)^2}$$

$$= \frac{960+30}{2008-225} = \frac{990}{1783}$$

$$= 0.555$$

Regression Coefficient of x on y is

$$b_{xy} = b_{uv} = \frac{n\Sigma uv - (\Sigma u)(\Sigma v)}{n\Sigma v^2 - (\Sigma v)^2}$$

$$= \frac{990}{8 \times 74 - 4} = 1.68$$

The equation of the line of regression of x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 158.13 = 1.68(y - 159.25)$$

$$x = 1.68y - 267.54 + 158.13$$

$$x = 1.68y - 109.41$$

The equation of the line of regression of y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 159.25 = 0.56(x - 158.13)$$

$$y = 0.56x - 88.55 + 159.25$$

$$y = 0.56x + 70.697.$$

When the height of the father (x) is 154 cm, the height of the son (y) we consider the equation of line of regression of y on x.

$$\text{ie } y = 0.56x + 70.697$$

$$= 0.56(154) + 70.697$$

$$y = 156.937$$

(3) The two lines of regression are

$$8x - 10y + 66 = 0 \longrightarrow (A)$$

$$40x - 18y - 214 = 0 \longrightarrow (B)$$

The variance of x is 9. Find

i) The mean values of x and y

ii) Correlation coefficient between x & y.

Sol:

(i) Since both the lines of regression passes through the mean values $\bar{x}$ and $\bar{y}$.

the point $(\bar{x}, \bar{y})$ must satisfy the two given regression lines.

ie $8\bar{x} - 10\bar{y} = -66 \longrightarrow (1)$

$40\bar{x} - 18\bar{y} = 214 \longrightarrow (2)$

$(1) \times 5 \Rightarrow 40\bar{x} - 50\bar{y} = -330$

$(2) \Rightarrow 40\bar{x} - 18\bar{y} = 214$

$(-) \quad (+) \quad (-)$

$-32\bar{y} = -544$

$\boxed{\bar{y} = 17}$

$8\bar{x} - 10(17) = -66$

$8\bar{x} = -66 + 170$

$8\bar{x} = 104$

$\boxed{\bar{x} = 13}$

Hence the mean values are given by $\bar{x} = 13$, $\bar{y} = 17$.

(ii)

Let us suppose that equation (A) is the equation of line of regression of y on x and (B) is the equation of line of regression of x on y we get, after rewritting (A) & (B),

$$10y = 8x + 66$$

$$y = \frac{8}{10}x + \frac{66}{10}$$

$$\therefore \boxed{b_{yx} = 8/10}$$

$$40x = 18y + 214$$

$$x = \frac{18}{40}y + \frac{214}{40}$$

$$\boxed{b_{xy} = \frac{18}{40}}$$

$\therefore r^2 = b_{xy} \times b_{yx}$

$$r^2 = \frac{18}{40} \times \frac{8}{10} = \frac{9}{25}$$

$$\therefore r = \pm \frac{3}{5} = \pm 0.6$$

Since both the regression coefficients are positive, r must be positive.

$\therefore r = 0.6$.

Note:

If we take equation (A) as the line of regression of x on y we get, $8x = 10y - 66 \Rightarrow x = \frac{10}{8}y - \frac{66}{8}$

and if we take equation (B) as the line of regression of y on x we get

$$18y = 40x - 214 \Rightarrow y = \frac{40}{18}y - \frac{214}{18}$$

$$b_{xy} = \frac{10}{8}, \quad b_{yx} = \frac{40}{18}$$

$$\therefore \quad r^2 = b_{xy}\, b_{yx} = \frac{10}{8} \times \frac{40}{18} = \frac{25}{9} = 2.78$$

But r^2 should always lies between 0 and 1.

Hence our assumption that line (A) is line of regression of x on y and the line (B) is line of regression of y on x is wrong.

(4) The following data were available $\bar{x} = 970$, $\bar{y} = 18$, $\sigma_x = 38$, $\sigma_y = 2$. Correlation Coefficient $r = 0.6$. Find the line of regression and obtain the value of x and $y = 20$.

Sol: We know that the line of regression of x on y is given by

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

Given $\bar{x} = 970$, $\bar{y} = 18$, $\sigma_x = 38$, $\sigma_y = 2$, $r = 0.6$

$$x - 970 = 0.6 \times \frac{38}{2}(y - 18)$$

$$= 11.4y - 205.2$$

$$x = 11.4y + 764.8$$

This gives the line of x on y.

Now when $y = 20$, we get

$$x = 11.4 \times 20 + 764.8$$

$$\boxed{x = 992.8}$$

(5) The regression equation of X and Y is $3y-5x+108=0$. If the mean value of Y is 44 and the Variance of X were $\frac{9}{16}$ th of the Variance of Y. Find the mean value of X and the Correlation Coefficient.

Sol:

Since the line of regression passes through $(\bar{x}, \bar{y})$, we get

$$3\bar{y} - 5\bar{x} + 108 = 0$$

$$3(44) - 5\bar{x} + 108 = 0$$

$$5\bar{x} = 108 + 132$$

$$5\bar{x} = 240$$

$$\boxed{\bar{x} = 48}$$

Mean value of X is 48.

Given $3y - 5x + 108 = 0$

$$5x = 3y + 108$$

$$x = \frac{3}{5} y + \frac{108}{5}$$

which is the line of regression of x on y.

$$\therefore b_{xy} = 3/5$$

ie $$r \frac{\sigma_x}{\sigma_y} = 3/5 \longrightarrow (1)$$

Given $$\sigma_x^2 = \frac{9}{16} \sigma_y^2$$

$$\sigma_x = \frac{3}{4} \sigma_y \longrightarrow (2)$$

(2) in (1)

$$r \frac{3}{4} \frac{\sigma_y}{\sigma_y} = 3/5$$

$$r = 12/15 = 0.8$$

$\therefore$ Correlation Coefficient is $r = 0.8$.

Multiple Correlation:

The value of one variable influenced by many other variables is known as multiple correlation.

If x_1 is the dependent variable & $x_2, x_3 \ldots x_n$ are independent variables then we donote the multiple correlation by $R_{1.23\ldots n}$.

Example:

The yield of crop (X_1) may depend upon the rainfall, (X_2) fertilizer, (X_3) the average temperature, (X_4) the period between sowing & harvesting. (X_5)

Partial Correlation:

Partial Correlation Coefficient measure the relationship between any two variables when the other variables connected with those variables are kept constant.

Example:

Correlation between the height & weight of the boys of the same age.

If x_1 & x_2 are partially correlated and x_3 kept as constant then we denote the partial correlation by $r_{12.3}$.

Formula:

Multiple Correlation:

If X_1 is correlated with X_2 & X_3 then the multiple correlation is

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1-r_{23}^2}}$$

|||ly

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1-r_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{r_{13}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1-r_{12}^2}}$$

Properties of multiple Correlation Coefficient:

1. $0 \le R_{1.23} \le 1$

2. $R_{1.23} \geq r_{12}, r_{13}, r_{23}$

Partial Correlation:

The partial Correlation Coefficient between X_1 & X_2 and X_3 kept as Constant is given by

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}}$$

$$r_{13.2} = \frac{r_{13} - r_{12} \cdot r_{23}}{\sqrt{(1-r_{12}^2)(1-r_{23}^2)}}$$

$$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{(1-r_{12}^2)(1-r_{13}^2)}}$$

(1) From the data relating to the yield of dry bark (X_1), height (X_2) & girth (X_3) for 18 cinchona plants the following correlation coefficients were obtained $r_{12} = 0.77$, $r_{13} = 0.72$ and $r_{23} = 0.52$. Find the partial correlation coefficient $r_{12.3}$, $r_{13.2}$, $r_{23.1}$ & multiple correlation coefficients $R_{1.23}$.

Solution:

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1-r_{13}^2}\sqrt{1-r_{23}^2}} = \frac{0.77 - 0.72 \times 0.52}{\sqrt{1-0.72^2}\sqrt{1-0.52^2}}$$

$$\boxed{r_{12.3} = 0.62}$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{23}}{\sqrt{1-r_{12}^2}\sqrt{1-r_{23}^2}}$$

$$r_{13.2} = \frac{0.72 - 0.77 \times 0.52}{\sqrt{1-0.77^2}\sqrt{1-0.52^2}} = 0.6866$$

$$\boxed{r_{13.2} = 0.6866}$$

$$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{1-r_{12}^2}\,\sqrt{1-r_{13}^2}}$$

$$= \frac{0.52 - 0.77 \times 0.72}{\sqrt{1-0.77^2}\,\sqrt{1-0.72^2}}$$

$$\boxed{r_{23.1} = -0.1119}$$

$$R^2_{1.23} = \frac{r_{12}^2 + r_{13}^2 - 2 r_{12} r_{13} r_{23}}{1-r_{23}^2}$$

$$= \frac{(0.77)^2 + (0.72)^2 - 2(0.77)(0.72)(0.52)}{1-(0.52)^2}$$

$$R^2_{1.23} = 0.7334$$

$$\boxed{R_{1.23} = 0.8564}$$

(Multiple Correlation Coefficient is non negative)

Regression Coefficient of $X_{1.3}$ on $X_{2.3}$

$$b_{12.3} = r_{12.3} \frac{\sigma_{1.3}}{\sigma_{2.3}}$$

where $r_{12.3} = \dfrac{r_{12} - r_{13} r_{23}}{\sqrt{1-r_{13}^2}\sqrt{1-r_{23}^2}}$

$$\sigma_{1.3} = \sigma_1 \sqrt{1-r_{13}^2}$$

$$\sigma_{2.3} = \sigma_2 \sqrt{1-r_{23}^2}$$

(or)

$$b_{12.3} = \frac{r_{12} - r_{13} r_{23}}{1-r_{23}^2} \frac{\sigma_1}{\sigma_2}$$

$|||^{ly}$

$$b_{13.2} = \frac{r_{13} - r_{23} r_{12}}{1-r_{23}^2} \frac{\sigma_1}{\sigma_3}$$

$$b_{13.2} = r_{13.2} \frac{\sigma_{1.2}}{\sigma_{3.2}}$$

where $r_{13.2} = \dfrac{r_{13} - r_{12} r_{32}}{\sqrt{1-r_{12}^2}\sqrt{1-r_{32}^2}}$

$$\sigma_{1.2} = \sigma_1 \sqrt{1-r_{12}^2}$$

$$\sigma_{3.2} = \sigma_3 \sqrt{1-r_{32}^2}$$

$$\sigma_{1.23} = \sigma_1 \sqrt{\frac{1 - r_{12}^2 - r_{13}^2 - r_{23}^2 + 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

(or)

$$\boxed{\sigma_{1.23} = \sigma_1 \sqrt{1 - R_{1.23}^2}}$$

① In a trivariate distribution $\sigma_1 = 2$; $\sigma_2 = \sigma_3 = 3$, $r_{12} = 0.7$, $r_{23} = r_{31} = 0.5$ then find i) $r_{23.1}$ ii) $R_{1.23}$ iii) $b_{12.3}$, $b_{13.2}$, $\sigma_{1.23}$.

Solution:

i) $$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{(1 - r_{12}^2)(1 - r_{13}^2)}}$$

$$= \frac{0.5 - 0.7 \times 0.5}{\sqrt{(1 - 0.49)(1 - 0.25)}} = 0.2425$$

ii) $$R_{1.23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}$$

$$= \frac{0.49 + 0.25 - 2(0.7)(0.5)(0.5)}{1 - 0.25} = 0.52$$

$\therefore R_{1.23} = 0.7211$

iii) $b_{12.3} = r_{12.3} \frac{\sigma_{1.3}}{\sigma_{2.3}}$ and $b_{13.2} = r_{13.2} \frac{\sigma_{1.2}}{\sigma_{3.2}}$

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}} = 0.6$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{32}}{\sqrt{(1-r_{12}^2)(1-r_{32}^2)}} = 0.2425$$

$$\sigma_{1.3} = \sigma_1 \sqrt{1-r_{13}^2} = 2\sqrt{1-0.25} = 1.7320$$

$$\sigma_{2.3} = \sigma_2 \sqrt{1-r_{23}^2} = 3\sqrt{1-0.25} = 2.5980$$

$$\sigma_{1.2} = \sigma_1 \sqrt{1-r_{12}^2} = 2\sqrt{1-0.49} = 1.4282$$

$$\sigma_{3.2} = \sigma_3 \sqrt{1-r_{32}^2} = 3\sqrt{1-0.25} = 2.5980$$

Hence $b_{12.3} = 0.4$ and $b_{13.2} = 0.1333$

iv) $\sigma_{1.23} = \sigma_1 \sqrt{1 - R_{1.23}^2}$

$= 2 \times \sqrt{1 - 0.52}$

$\boxed{\sigma_{1.23} = 1.3856}$